插图英语阅读小册

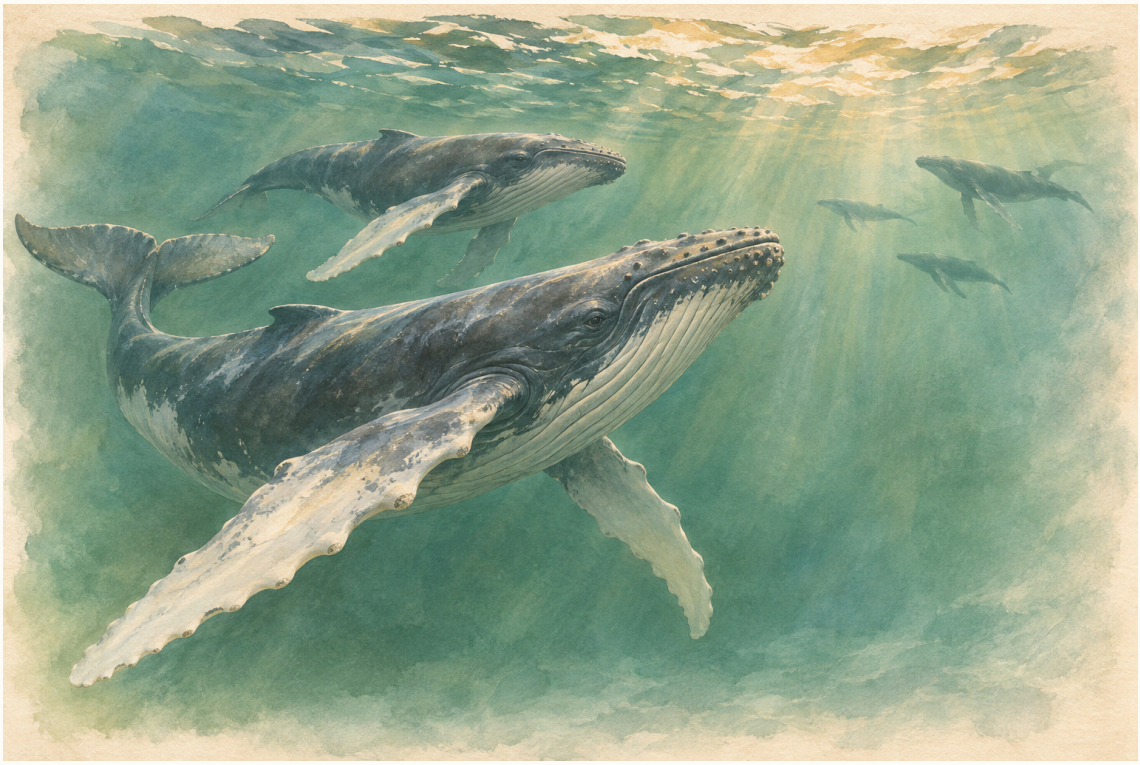
第7册 · 十篇故事

按推荐顺序阅读，保留原篇号。点击篇目即可跳转；每篇依次包含插图、英文原文、中文译文、词汇与回想。

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鲸歌易帜

A Change of Anthem in the Deep



新歌如何跨过海洋，传遍鲸群？

英文原文

Each winter, male humpback whales, and only males, fill their breeding grounds with a haunting melody. Sound units constitute phrases, and phrases build themes in an elaborate hierarchy. Each rendition lasts minutes, and the song can recur for hours, probably for courtship. All the males in a population sing an identical version, which evolves subtly each year.

That is the rule. Off eastern Australia, it was broken. From 1995 to 1998, researchers used hydrophones, or underwater microphones, to record humpbacks on their seasonal migration along the coastline. In the mid-1990s, merely two of 82 singers emitted a distinctive new song, utterly unlike the reigning chorus.

Within about two years, virtually every singer had converted, and by 1998 the old song had been displaced wholesale. A study published in Nature in November 2000 found that the new song resembled that of the Indian Ocean population off western Australia, apparently ferried east by a handful of visiting performers.

Nor was it a freak. In April 2011, a study in Current Biology analysed over a decade of recordings from six South Pacific populations. Song types primarily arose off eastern Australia, then rippled eastward through successive populations to French Polynesia, generally crossing the region within about two years.

Such rapid change, too fast for genetic evolution, points to imitation rather than inheritance, a persuasive case of cultural transmission. Why males sing, and why they overhaul their songs, remain controversial. One idea is that novelty itself spurs change. Some scholars are reluctant to define this as culture in the human sense. Whatever the label, a few outsiders off eastern Australia reshaped an entire population's music.

中文译文

每到冬季，雄性座头鲸——而且只有雄性——会让繁殖地回荡着令人难忘的旋律。声音单元组成乐句，乐句再构成主题，形成复杂精细的层级结构。每次歌唱持续数分钟，又能连续重复数小时，很可能是为了求偶。同一种群中的所有雄性都唱着相同版本，这个版本每年都会发生细微变化。

这是通常的规律，但在澳大利亚东部海域，规律被打破了。1995至1998年，研究者用水听器，也就是水下麦克风，记录沿岸季节性迁徙的座头鲸。20世纪90年代中期录到的82头歌者中，只有两头唱出了一首全新的歌，与当时占主流的合唱完全不同。

大约两年内，几乎每头歌者都换了歌；到1998年，旧歌已被彻底取代。2000年11月发表于《自然》的一项研究发现，新歌与澳大利亚西部近海的印度洋种群所唱的歌相似，似乎是少数来访歌者把它向东带来的。

这也并非孤例。2011年4月，《当代生物学》发表的一项研究分析了南太平洋六个种群十余年的录音。歌型大多起源于澳大利亚东部海域，随后依次经由一个个种群向东传播，直到法属波利尼西亚，通常约两年就能横跨这片区域。

这种变化快得难以用基因演化解释，指向的是模仿而非遗传，为文化传递提供了一个有说服力的例证。但雄性为何歌唱、为何全面更换歌曲，仍有争议。一种观点认为，新奇本身就会推动变化。有些学者不愿把它定义为人类意义上的文化。无论采用什么名称，澳大利亚东部海域的少数外来者，确实重塑了整个种群的音乐。

核心词汇

先记这22项。连着下方的原文短语一起回想。

constitute v.
构成
Sound units constitute phrases,

elaborate adj.
复杂精细的
themes in an elaborate hierarchy.

hierarchy n.
层级结构
themes in an elaborate hierarchy.

recur v.
反复出现
and the song can recur for hours,

courtship n.
求偶
for hours, probably for courtship.

identical adj.
完全相同的
population sing an identical version,

migration n.
迁徙
on their seasonal migration along the

distinctive adj.
与众不同的
emitted a distinctive new song,

virtually adv.
几乎
two years, virtually every singer had

displace v.
取代
old song had been displaced wholesale.

resemble v.
与……相似
the new song resembled that of the

a handful of n.phr.
少数；一小批
by a handful of visiting performers.

primarily adv.
主要地
Song types primarily arose off eastern

successive adj.
相继的
through successive populations to

evolution n.
演化
too fast for genetic evolution,

imitation n.
模仿
to imitation rather than inheritance,

inheritance n.
遗传
to imitation rather than inheritance,

transmission n.
传递；传播
case of cultural transmission.

overhaul v.
全面改造
why they overhaul their songs, remain

controversial adj.
存在争议的
their songs, remain controversial.

novelty n.
新奇
is that novelty itself spurs change.

reluctant adj.
不情愿的
Some scholars are reluctant to define

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

male adj.
雄性的

humpback whale n.phr.
座头鲸

breeding ground n.phr.
繁殖地

haunting adj.
萦绕心头的；难忘的

melody n.
旋律

sound unit n.phr.
声音单元

theme n.
主题；主题段落

rendition n.
一次演唱；演绎

population n.
生物种群

evolve v.
演变

subtly adv.
细微地；不易察觉地

off prep.
在……近海

researcher n.
研究者

microphone n.
麦克风

seasonal adj.
季节性的

coastline n.
海岸线

merely adv.
仅仅

emit v.
发出

utterly unlike adj.phr.
完全不同于

reigning adj.
当前占主导的

chorus n.
合唱

convert v.
转变；此处指改唱

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

wholesale adv.
全面地；全盘地

ferry v.
带来；运送

performer n.
表演者；此处指歌者

freak n.
罕见的例外

analyse v.
分析

recording n.
录音

Pacific proper n.
太平洋

ripple v.
如涟漪般扩散

eastward adv.
向东

Polynesia proper n.
波利尼西亚

generally adv.
通常

region n.
区域

rapid change n.phr.
快速变化

persuasive adj.
有说服力的

cultural adj.
文化的

spur v.
推动；促进

scholar n.
学者

define as phr.v.
与as连用：把……界定为

as prep.
标示被归入的类别

outsider n.
外来者

reshape v.
重塑

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Sound units _____ phrases,

02 _____ every singer had converted,

03 _____ visiting performers.

04 Some scholars are _____ to define this as culture in the human sense.

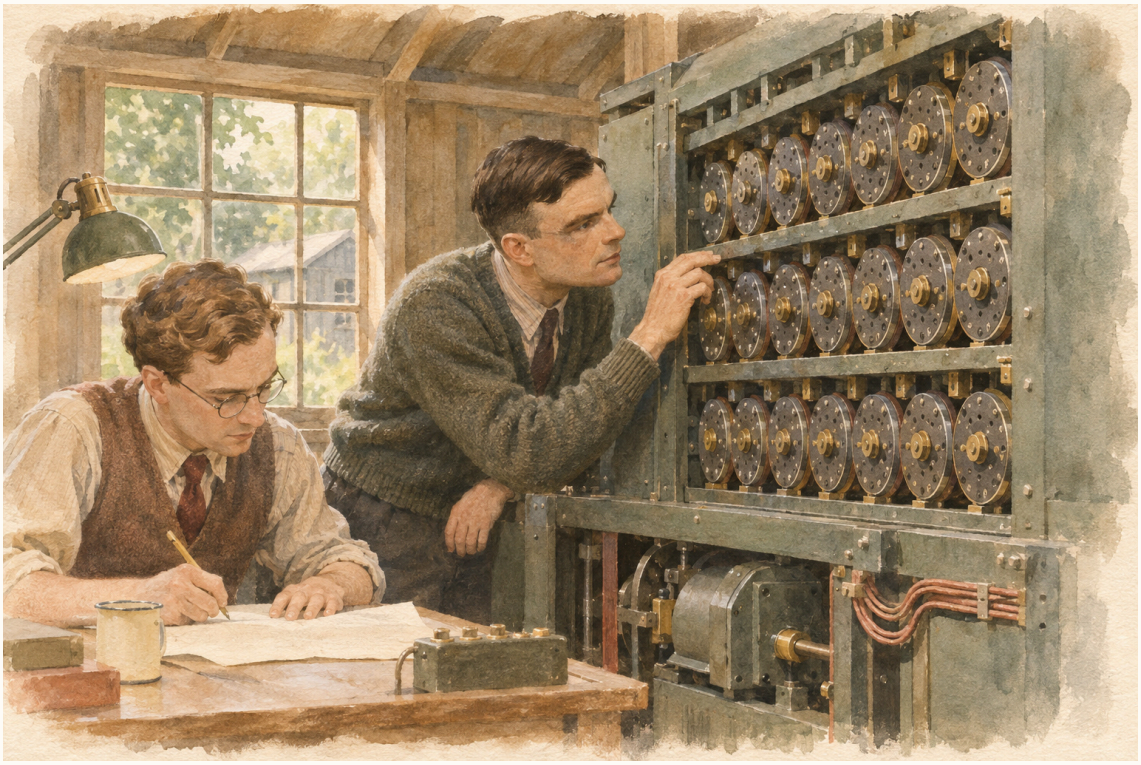
文章主线

- 常态 同一种群的雄性座头鲸唱相同版本的歌，通常每年只作细微变化，歌唱可能用于求偶。
- 换歌 澳洲东部录到的82头鲸中，最初仅两头唱全新歌；约两年后，几乎整个种群都改唱新歌。
- 来源 新歌与澳洲西部种群的歌相似，研究推测是少量来访鲸把它带到了东部。
- 传播 六个南太平洋种群的十余年录音显示，歌型主要从澳洲东部逐群向东扩散，通常约两年传到法属波利尼西亚。
- 解释 变化太快，支持模仿而非遗传的解释；为何换歌、是否等同人类文化，仍有争议。

参考答案　1 constitute 2 virtually 3 a handful of 4 reluctant

为机器点燃思想的人

The Man Who Gave Machines a Mind



图灵的贡献，为何没能保护他?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Born in June 1912, Alan Turing had a formidable intellect by his early twenties. In a paper read in November 1936, he outlined a "universal machine": a device able to execute any computation, given the right instructions. That abstract construct, rooted in mathematical logic, became the theoretical blueprint of the modern computer.

On 4 September 1939 he joined the classified codebreaking effort at Bletchley Park, leading work on naval Enigma, Germany's encryption machine. Polish mathematicians had deciphered Enigma by late 1932 and shared their insights with Britain and France in July 1939. His team's machines exploited them and, operational from March 1940, automated the laborious search for each day's rotor settings.

In October 1950, writing in the philosophy journal Mind, he posed a provocative question: "Can machines think?" He proposed a practical substitute, the imitation game, now widely called the Turing test. An interrogator exchanges typed messages with two concealed players, one human and one machine. If the interrogator cannot consistently distinguish them, the machine passes.

His exceptional wartime contribution did not protect him. On 31 March 1952 he was convicted of "gross indecency", the charge then used against gay men. He forfeited his security clearance and, to avert imprisonment, submitted to hormone injections. He died of cyanide poisoning on 7 June 1954. A coroner's inquest ruled it suicide, though some doubt it.

Recognition lagged. On 10 September 2009 Britain's prime minister publicly apologised, and on 24 December 2013 Turing was posthumously granted a royal pardon. On 23 June 2021, his birthday, a fifty-pound note bearing his portrait entered circulation. The country that prosecuted him put his face on its money.

中文译文

艾伦·图灵生于1912年6月，二十出头便展现出非凡才智。1936年11月宣读的一篇论文中，他提出“通用机器”的轮廓：只要给出正确指令，这种装置就能执行任何计算。这一植根于数理逻辑的抽象构想，后来成为现代计算机的理论蓝图。

1939年9月4日，图灵加入布莱切利园的机密破译工作，领导针对德国海军恩尼格玛加密机的研究。波兰数学家早在1932年底就已破译恩尼格玛，并于1939年7月向英国和法国分享了思路。图灵团队的机器借鉴这些成果，从1940年3月投入运行，将每天搜寻转子设置的繁重工作自动化。

1950年10月，图灵在哲学期刊《心灵》上提出一个发人深思的问题：“机器能思考吗？”他建议用一种实际可行的替代办法来讨论这个问题，即“模仿游戏”，如今通常称为图灵测试。提问者通过打字与两名隐藏的参与者交流，一方是人，另一方是机器。如果提问者无法稳定地区分二者，机器就算通过测试。

卓越的战时贡献并未保护图灵。1952年3月31日，他因“严重猥亵”罪被定罪，这是当时用来惩治男同性恋者的罪名。他失去接触机密资料的资格，并为避免入狱而接受激素注射。1954年6月7日，他死于氰化物中毒。验尸官的死因调查裁定为自杀，但有人对此存疑。

认可来得很迟。2009年9月10日，英国首相公开道歉；2013年12月24日，图灵获得身后皇家赦免。2021年6月23日，他生日当天，印有他肖像的50英镑纸币开始流通。曾经起诉他的国家，将他的面容印在了自己的货币上。

图灵测试是行为区分标准，不能直接等同于机器具有意识的证明。

核心词汇

先记这22项。连着下方的原文短语一起回想。

intellect n.
才智；智力
a formidable intellect

execute v.
执行（指令或任务）
able to execute any computation

computation n.
计算
able to execute any computation

abstract adj.
抽象的
That abstract construct

construct n.
理论构想；概念模型
That abstract construct

blueprint n.
蓝图；设计基础
the theoretical blueprint

classified adj.
机密的；保密的
the classified codebreaking effort

encryption n.
加密
Germany's encryption machine

decipher v.
破译；解读
had deciphered Enigma

insight n.
深刻见解；关键思路
shared their insights

operational adj.
投入运行的
operational from March 1940

automate v.
使自动化
automated the laborious search

laborious adj.
费力耗时的
automated the laborious search

provocative adj.
发人深思的；引起争议的
a provocative question

substitute n.
替代办法；替代物
a practical substitute

consistently adv.
稳定地；始终如一地
cannot consistently distinguish them

distinguish v.
区分；辨别
cannot consistently distinguish them

be convicted of 动词短语
被判犯有……罪
he was convicted of

forfeit v.
丧失；被剥夺
He forfeited his security clearance

avert v.
避免；防止
to avert imprisonment

posthumously adv.
在死后；身后地
posthumously granted

pardon n.
赦免
a royal pardon

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

formidable adj.
非凡的；强大的

outline v.
概述；勾勒

device n.
装置；设备

be rooted in 动词短语
根源于；植根于

mathematical adj.
数学的

logic n.
逻辑

theoretical adj.
理论的

codebreaking effort 名词短语
密码破译工作

naval adj.
海军的

exploit v.
利用；运用

rotor n.
转子

philosophy n.
哲学

pose v.
提出（问题）

propose v.
提议；建议

practical adj.
切实可行的

widely adv.
广泛地；普遍地

interrogator n.
提问者；询问者

concealed adj.
隐藏的

exceptional adj.
杰出的；非凡的

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

wartime adj.
战时的

contribution n.
贡献

security clearance 名词短语
涉密资格；接触机密的许可

imprisonment n.
监禁；入狱

submit to 动词短语
接受；屈从于

hormone n.
激素

injection n.
注射

cyanide poisoning 名词短语
氰化物中毒

coroner n.
验尸官

suicide n.
自杀

recognition n.
认可；承认

lag v.
滞后；迟迟未到

publicly adv.
公开地

apologise v.
道歉

be aux.
构成过去时被动语态

grant v.
授予；给予

portrait n.
肖像

circulation n.
（货币等的）流通

prosecute v.
起诉；提起公诉

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 That abstract construct, rooted in mathematical logic, became the theoretical _____ of the modern computer.

02 His team's machines exploited them and, operational from March 1940, automated the _____ search for each day's rotor settings.

03 If the interrogator cannot consistently _____ them, the machine passes.

04 Turing was _____ granted a royal pardon.

文章主线

- 理论 1936年，图灵提出按指令执行各种计算的通用机器模型，为现代计算机奠定理论基础。
- 破译 图灵团队借鉴波兰的破译成果，1940年用机器自动搜索德国海军密码机每天的设置，加速破译。
- 测试 1950年，他提出模仿游戏：通过打字交谈，提问者若不能稳定分清人和机器，机器就通过测试。
- 迫害 1952年图灵因同性恋行为获罪，被取消涉密资格，为免入狱接受激素注射；1954年死于氰化物中毒。
- 追认 其死被判为自杀，但有人质疑。英国2009年道歉、2013年给予身后赦免；2021年肖像登上50镑钞票。

参考答案　1 blueprint 2 laborious 3 distinguish 4 posthumously

没有到来的零日

The Day Zero That Never Came



开普敦怎样避开了水龙头大范围关闭的零日？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In January 2018, Cape Town issued a stark alert: Day Zero could arrive in April. The term denoted a grim threshold, reached when water in the city's reservoirs, as a percentage of capacity, fell to 13.5 per cent. Most taps would then be shut, and residents would queue at about 200 designated collection points.

The prospect was genuine. Three consecutive years of meagre rainfall, from 2015 to 2017, had diminished the reservoirs serving a metropolitan area of about four million people. By early 2018 the dams were roughly a fifth full. The countdown turned a chronic shortage into an imminent deadline.

The city paired mandatory restrictions with unprecedented transparency. From 1 February 2018, each person was allotted 50 litres a day. Officials disclosed dam levels and household consumption; the published tallies made thrift visible and let neighbours benchmark their use. Daily use was more than halved, from about 1.2 billion litres in 2015 to about 500 million by early 2018.

Other forces intervened. In February 2018, upstream farmers released water from their irrigation dams; with farms drawing less, Day Zero was postponed to July. In March 2018, officials judged Day Zero unlikely that year. Winter rains subsequently replenished the reservoirs, which were about half full by July. The 50-litre limit was eased in October 2018.

The burden, critics argued, fell unevenly. Townships, poor areas segregated by race under apartheid, had long made do with scant water, yet steeper tariffs squeezed them hardest. Others blamed alarmist warnings for tourist cancellations. Households, farmers, and rain kept the collection points idle, but the sacrifice weighed heaviest on those with the least to give.

中文译文

2018年1月，开普敦发出严峻警报：“零日”可能在4月到来。这个词指一道令人忧惧的门槛：城市各水库蓄水量降至总容量的13.5%。届时，大多数水龙头将关闭，居民必须到约200个指定取水点排队领水。

这并非虚构的危险。2015至2017年连续三年降雨稀少，为约400万人口的都会区供水的水库蓄水持续下降。到2018年初，蓄水量只剩约五分之一。倒计时把长期缺水变成了一个迫在眉睫的最后期限。

市政府把强制限水与前所未有的信息公开结合起来。从2018年2月1日起，每人每天限用50升水。官员公布水库水位和家庭用水量，让节水成效看得见，也让邻居能相互比较用水情况。全市日用水量已从2015年的约12亿升降至2018年初的约5亿升，减少了一半以上。

其他因素也发挥了作用。2018年2月，上游农民从自有灌溉水库放水，再加上农场减少取水，“零日”被推迟至7月。到了3月，官员判断当年不大可能出现“零日”。随后到来的冬雨补充了水库，到7月时蓄水已恢复至约半满。2018年10月，每人每日50升的限制得到放宽。

批评者指出，负担并不均等。种族隔离时期按种族划出的贫困聚居区，长期以来就只能用少量水勉强生活，却最难承受提高后的水价。也有人指责，危言耸听的警报导致游客取消行程。居民、农民和雨水共同让取水点没有派上用场，但付出最沉重代价的，却是那些最没有余力付出的人。

核心词汇

先记这22项。连着下方的原文短语一起回想。

issue v.
发出；发布
Cape Town issued a stark alert

stark adj.
严峻鲜明的
a stark alert

denote v.
指；表示
The term denoted a grim threshold

threshold n.
门槛；临界点
a grim threshold

reservoir n.
水库
the city's reservoirs

capacity n.
容量
as a percentage of capacity

designate v.
指定
200 designated collection points

consecutive adj.
连续的
Three consecutive years

meagre adj.
少得可怜的；不足的
meagre rainfall

chronic adj.
长期持续的
a chronic shortage

imminent adj.
迫在眉睫的
an imminent deadline

mandatory adj.
强制性的
mandatory restrictions

unprecedented adj.
前所未有的
unprecedented transparency

transparency n.
公开透明
unprecedented transparency

allot v.
分配；配给
each person was allotted 50 litres

consumption n.
消耗量；用水量
household consumption

benchmark v.
与基准或他人比较
benchmark their use

postpone v.
推迟
Day Zero was postponed to July

replenish v.
补充；重新补给
replenished the reservoirs

burden n.
负担
The burden, critics argued

unevenly adv.
不均等地
fell unevenly

tariff n.
水价；收费标准
steeper tariffs squeezed them hardest

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

alert n.
警报

grim adj.
严酷的；令人忧惧的

percentage n.
百分比

resident n.
居民

prospect n.
可能出现的局面

genuine adj.
真实的；确实存在的

diminish v.
减少；使减少

metropolitan adj.
大都市的

dam n.
水坝；此处指水库

roughly adv.
大约

countdown n.
倒计时

shortage n.
短缺

restriction n.
限制措施

litre n.
升

official n.
官员

disclose v.
公布；披露

household n.
家庭

tally n.
统计数目

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

thrift n.
节约；节俭

visible adj.
看得见的

halve v.
减半

intervene v.
介入；发挥作用

upstream adj.
上游的

irrigation n.
灌溉

unlikely adj.
不大可能的

subsequently adv.
随后

ease v.
放宽；减轻

township n.
隔离制下的贫困聚居区

segregate v.
按种族隔离

apartheid n.
种族隔离制度

scant adj.
很少的；不足的

steep adj.
价格更高的

squeeze v.
使承受经济压力

alarmist adj.
危言耸听的

cancellation n.
取消；退订

idle adj.
闲置的

sacrifice n.
牺牲；付出的代价

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The term denoted a grim _____,

02 The city paired _____ restrictions with unprecedented transparency.

03 each person was _____ 50 litres a day.

04 Winter rains subsequently _____ the reservoirs, which were about half full by July.

文章主线

- 危机 连续三年少雨使开普敦水库告急，2018年初当局警告，四月可能迎来零日。
- 门槛 零日指蓄水降至容量的13.5%，届时大部分水龙头关闭，居民须到指定地点领水。
- 节水 日用水量由2015年约12亿升降至2018年初约5亿升；2月起限每人每日50升，并公开水位及用量。
- 转机 居民节水、农场放水及减少取水，使零日一再推迟；随后冬雨补充水库，避免了当年大范围停水。
- 负担 批评者指出，贫困社区原本用水就少，却最难承受上涨水价，节水代价并不均等。

参考答案　1 threshold 2 mandatory 3 allotted 4 replenished

改变人类命运的霉菌

The Mould That Changed the World



从一团霉菌到救命药，中间还差什么？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In September 1928 Alexander Fleming, a bacteriologist at St Mary's Hospital, London, returned from holiday to find that a stray mould had contaminated a dish of staphylococcus bacteria. Nothing grew around it. Popular mythology blames an open window; the spores probably originated in a laboratory downstairs, where a colleague cultivated moulds.

Fleming called the mould's juice penicillin and published in June 1929. Yet the crude juice was unstable and stubbornly hard to purify, and he and the chemists he enlisted lacked the expertise to refine it. He used it chiefly to isolate other microbes, and penicillin lay dormant, a curiosity rather than a cure.

A decade later, in Oxford, Howard Florey, an Australian pathologist, revived the work with Ernst Chain, a German-born chemist and Jewish refugee. Their colleague Norman Heatley improvised an ingenious apparatus to extract the compound. On 25 May 1940 eight mice received a lethal dose of bacteria; only the four also injected with penicillin lived. The Lancet, a prestigious journal, printed the outcome that August.

In February 1941 Albert Alexander, an Oxford constable gravely ill with infection, rallied on penicillin. When the scarce supply ran out, he died in March: the obstacle was quantity. From July 1941 a government laboratory in Peoria, Illinois, boosted yields about tenfold with corn-steep liquor, a starch by-product; later a mouldy market melon furnished the laboratory with a more prolific strain. By D-Day in June 1944 industrial fermentation met Allied needs.

In October 1945 Fleming, Florey, and Chain shared the Nobel prize; Heatley was overlooked. Fortune put the mould in Fleming's dish; tenacious teams and years of toil transformed it into a remedy.

中文译文

1928年9月，伦敦圣玛丽医院的细菌学家亚历山大·弗莱明度假回来，发现一团偶然出现的霉菌污染了一只培养葡萄球菌的培养皿，霉菌周围没有细菌生长。流行传说认为孢子从敞开的窗户飘入；它们其实很可能来自楼下的实验室，那里有位同事在培养霉菌。

弗莱明将霉菌产生的液汁称为青霉素，并于1929年6月发表研究。然而，粗制液汁不稳定，极难提纯；他和请来的化学家又缺乏精炼它的专业知识。他主要用它分离其他微生物，青霉素因此沉寂下来，仍是一件引人好奇的科学奇物，而非治病良药。

约十年后，在牛津，澳大利亚病理学家霍华德·弗洛里与德国出生的化学家、犹太难民恩斯特·钱恩重启研究。同事诺曼·希特利就地取材，制作了一套巧妙装置来提取这种化合物。1940年5月25日，八只小鼠接受致死剂量的细菌，只有同时注射青霉素的四只存活。权威期刊《柳叶刀》于同年8月发表了结果。

1941年2月，牛津警员艾伯特·亚历山大因感染病情危重，用上青霉素后有所好转。但稀缺的药物耗尽后，他于3月去世：难关在于产量。从1941年7月起，伊利诺伊州皮奥里亚的一家政府实验室利用淀粉生产副产物玉米浆，使产量达到原先的约十倍。后来，市场上的一只发霉甜瓜又提供了更高产的菌株。到1944年6月诺曼底登陆时，工业发酵已能满足盟军需求。

1945年10月，弗莱明、弗洛里和钱恩共同获得诺贝尔奖，希特利却被忽略了。机缘将霉菌送入弗莱明的培养皿，而多支团队锲而不舍、辛劳多年，才把它变成救人的良药。

核心词汇

先记这22项。连着下方的原文短语一起回想。

contaminate v.
污染；使混入杂质
a stray mould had contaminated

spore n.
孢子
the spores probably originated

penicillin n.
青霉素
the mould's juice penicillin

crude adj.
粗制的；未经精炼的
the crude juice

unstable adj.
不稳定的
the crude juice was unstable

purify v.
提纯；纯化
stubbornly hard to purify

expertise n.
专业知识；专长
lacked the expertise to refine it

isolate v.
分离；隔离
to isolate other microbes

revive v.
重启；使恢复活力
revived the work

improvise v.
就地取材制作；临时拼装
improvised an ingenious apparatus

ingenious adj.
巧妙的；独创的
an ingenious apparatus

apparatus n.
装置；设备
an ingenious apparatus

extract v.
提取
to extract the compound

lethal adj.
致死的
a lethal dose of bacteria

scarce adj.
稀缺的；不足的
the scarce supply

obstacle n.
障碍；难关
the obstacle was quantity

quantity n.
数量；产量
the obstacle was quantity

tenfold adv.
增至十倍
boosted yields about tenfold

prolific adj.
高产的；多产的
a more prolific strain

fermentation n.
发酵
industrial fermentation

tenacious adj.
锲而不舍的；顽强的
tenacious teams

remedy n.
良药；治疗手段
transformed it into a remedy

文中词汇与搭配

补充词汇共39项。保留每个表达在本文中的意思。

stray adj.
偶然出现的；零散的

mould n.
霉菌

bacterium n.
细菌（复数）

popular mythology 名词短语
流行传说

originate in 动词短语
起源于；来自

cultivate v.
培养；培育

stubbornly adv.
顽固地；格外难以

refine v.
精炼；提纯

chiefly adv.
主要地

microbe n.
微生物

lie dormant 动词短语
沉寂；暂未得到发展

curiosity n.
引人好奇的事物

cure n.
疗法；有效的治疗手段

chemist n.
化学家

refugee n.
难民

compound n.
化合物

dose n.
剂量

inject with 动词短语
注射（药物）

prestigious adj.
享有盛誉的

outcome n.
结果

文中词汇与搭配 续

补充词汇共39项。保留每个表达在本文中的意思。

constable n.
警员

gravely ill 形容词短语
病情危重的

infection n.
感染

rally v.
病情好转；恢复

run out 动词短语
用尽；耗尽

boost v.
提高；增加

corn-steep liquor 名词短语
玉米浆

starch n.
淀粉

mouldy adj.
发霉的

melon n.
甜瓜

furnish v.
提供；供应

with prep.
表示提供的东西

industrial adj.
工业的

Allied adj.
盟军的

overlook v.
忽视；未给予应有认可

fortune n.
机缘；运气

toil n.
辛劳；苦干

transform v.
使转化；改变

into prep.
变成；成为

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Yet the crude juice was unstable and stubbornly hard to _____, and he and the chemists he enlisted lacked the expertise to refine it.

02 Their colleague Norman Heatley _____ an ingenious apparatus to extract the compound.

03 When the scarce supply ran out, he died in March: the obstacle was _____.

04 By D-Day in June 1944 industrial _____ met Allied needs.

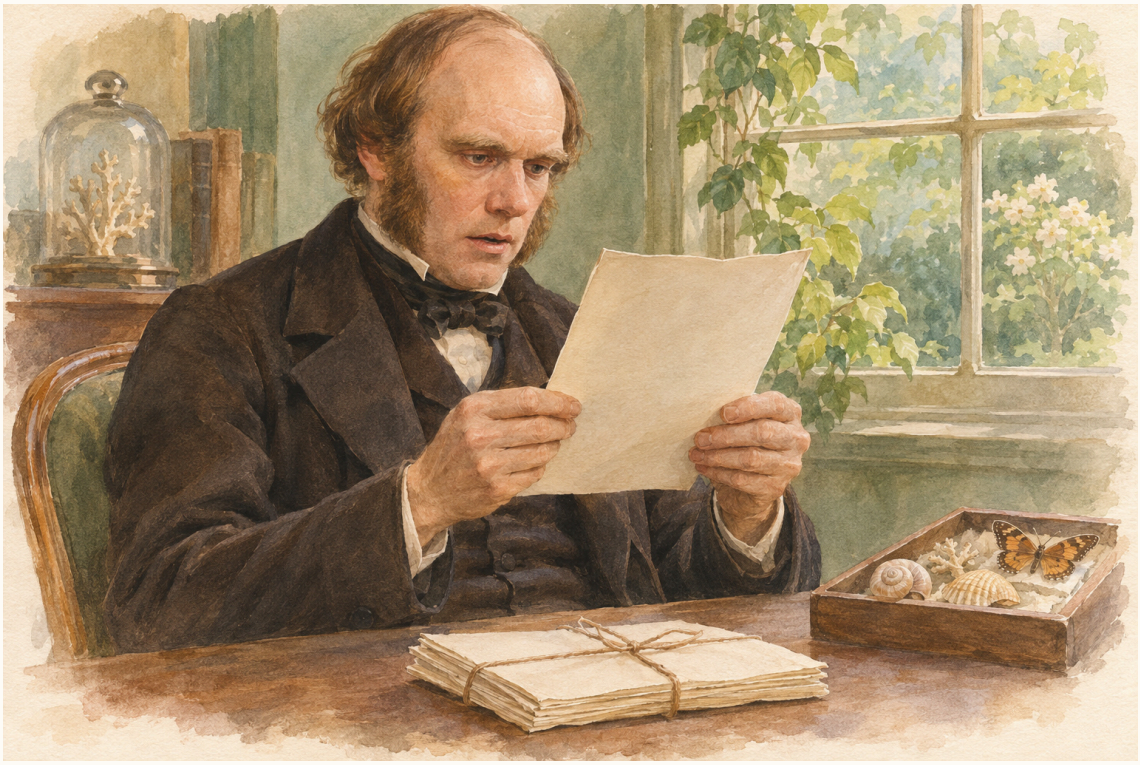
文章主线

- 偶然 1928年，弗莱明发现霉菌抑制细菌，把霉菌产物称作青霉素；孢子很可能来自楼下实验室，开窗说只是传闻。
- 难关 粗制青霉素不稳定、难提纯，最初多用于分离微生物；约十年后，弗洛里、钱恩和希特利等接力研究。
- 验证 8只感染小鼠中，只有用药的4只存活；1941年一名警员用药好转，却因药物耗尽死亡：瓶颈是产量。
- 量产 玉米浆使产量提高约十倍，发霉甜瓜又提供高产菌株；到1944年，工业发酵已能满足盟军用药。
- 归功 1945年弗莱明、弗洛里和钱恩获诺奖，希特利未入列；偶然发现经过多支团队多年攻关，才变成良药。

参考答案　1 purify 2 improvised 3 quantity 4 fermentation

一封信惊醒了达尔文

The Letter That Startled Darwin



华莱士的来信为何让达尔文不安?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Charles Darwin disembarked from HMS Beagle in October 1836. In September 1838 he began formulating the principle of natural selection: individuals better adapted to their surroundings survive and pass on their hereditary traits. By 1844 he had composed a 230-page manuscript, yet kept it unpublished, cautiously amassing evidence.

In February 1858, two decades after Darwin's first insight, Alfred Russel Wallace, a naturalist afflicted by fever in the Malay Archipelago, independently conceived the same idea. His concise essay reached Darwin on 18 June 1858. Its argument paralleled Darwin's own almost precisely; he told Charles Lyell he had never seen a more remarkable coincidence. His claim to priority seemed threatened.

Lyell and the botanist Joseph Hooker arranged for both papers to be read simultaneously at the Linnean Society, a scientific institution in London, on 1 July 1858. Neither author attended the proceedings. Wallace was abroad, and Darwin was grieving: scarlet fever had killed his infant son on 28 June. The compromise, struck without Wallace, may have kept rivalry over precedence at bay.

The papers' publication on 20 August 1858 met a muted reception. In May 1859 the Society's president remarked in his yearly address that the year had been barren of revolutionary discoveries. The sensation came on 24 November 1859 with Darwin's treatise, On the Origin of Species: booksellers had subscribed for every copy of its first impression, 1,250 in all.

Wallace had readily signified his acceptance in October 1858, assuring Hooker in a cordial letter of his satisfaction with the arrangement. In 1889 he published a lengthy exposition of the idea and entitled it Darwinism. Wallace never named the theory after himself.

中文译文

1836年10月，查尔斯·达尔文从小猎犬号下船。1838年9月，他开始形成自然选择原理：更能适应环境的个体存活下来，把遗传性状传给后代。到1844年，他已写出230页手稿，却没有发表，仍谨慎地积累证据。

1858年2月，也就是达尔文最初产生这个想法约二十年后，博物学家阿尔弗雷德·拉塞尔·华莱士在马来群岛身患热病时，独立想到了同一原理。他的简短论文于1858年6月18日送达达尔文手中，论证与达尔文自己的几乎完全一致。达尔文告诉查尔斯·莱尔，他从未见过如此惊人的巧合。他对首创地位的主张，似乎受到了威胁。

莱尔和植物学家约瑟夫·胡克安排两人的论文于1858年7月1日在伦敦的科学机构林奈学会上同场宣读。两位作者都没有到场：华莱士身在海外，达尔文则正沉浸在悲痛中——6月28日，猩红热夺走了他幼子的生命。这个在华莱士未参与商定的情况下达成的折中办法，或许避免了一场首创权之争。

论文于1858年8月20日发表，反响平淡。1859年5月，学会会长甚至在年度致辞中说，这一年没有革命性的发现。真正的轰动直到1859年11月24日达尔文的专著《物种起源》出版时才到来：书商已把首印的1250册全部订走。

华莱士已于1858年10月欣然表示接受这一安排，并在写给胡克的亲切信件中说明自己对此感到满意。1889年，他又出版了一部详细阐述这一思想的著作，题名《达尔文主义》。他从未用自己的名字为这套理论命名。

核心词汇

先记这22项。连着下方的原文短语一起回想。

formulate v.
形成并表述
he began formulating the principle of

principle n.
原理；原则
he began formulating the principle of

adapted to adj.phr.
适应……的
individuals better adapted to their

hereditary adj.
遗传的
and pass on their hereditary traits.

trait n.
性状；特征
and pass on their hereditary traits.

manuscript n.
手稿
he had composed a 230-page manuscript,

amass v.
积累；大量收集
it unpublished, cautiously amassing

afflicted by adj.phr.
遭受……折磨的
a naturalist afflicted by fever in the

independently adv.
独立地
independently conceived the same idea.

conceive v.
构想出
independently conceived the same idea.

concise adj.
简明的
His concise essay reached Darwin on 18

coincidence n.
巧合
seen a more remarkable coincidence.

priority n.
首创地位；优先权
claim to priority seemed threatened.

simultaneously adv.
同时地
to be read simultaneously at the

compromise n.
折中方案
compromise, struck without Wallace,

rivalry n.
竞争
may have kept rivalry over precedence

precedence n.
领先地位；先后次序
may have kept rivalry over precedence

muted adj.
平淡的；不热烈的
20 August 1858 met a muted reception.

reception n.
反响；接受情况
20 August 1858 met a muted reception.

cordial adj.
亲切友好的
a cordial letter of his satisfaction

exposition n.
详细阐述
a lengthy exposition of the idea and

entitle v.
为……题名
exposition of the idea and entitled it

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

disembark v.
下船；登岸

selection n.
选择；筛选

surroundings n.
周围环境

pass on phr.v.
传递；传给后代

compose v.
撰写

unpublished adj.
未发表的

cautiously adv.
谨慎地

naturalist n.
博物学家

parallel v.
与……十分相似

precisely adv.
精确地；完全地

remarkable adj.
非同寻常的

threaten v.
威胁

botanist n.
植物学家

arrange for phr.v.
安排……发生

institution n.
机构

proceedings n.
会议；会议活动

grieve v.
悲痛；哀悼

scarlet adj.
猩红色的；构成猩红热一词

infant adj.
婴幼儿的

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

publication n.
发表；出版

remark v.
评论；说

yearly adj.
年度的

barren of adj.phr.
缺乏；毫无

revolutionary adj.
革命性的

sensation n.
轰动

treatise n.
专著

On the Origin of Species proper n.
《物种起源》

subscribe for phr.v.
预订；订购

impression n.
印次；一次印刷

readily adv.
欣然地；毫不迟疑地

signify v.
表示；表明

acceptance n.
接受

assure v.
向……保证

satisfaction n.
满意

arrangement n.
安排

lengthy adj.
篇幅很长的

name after phr.v.
与after连用：以……命名

after prep.
标示命名所依据的人

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 cautiously _____ evidence.

02 His claim to _____ seemed threatened.

03 met a _____ reception.

04 assuring Hooker in a _____ letter

文章主线

- 酝酿 达尔文1838年构想自然选择：更适应环境者存活并传下遗传性状；1844年已写成230页手稿，仍搜证未发表。
- 来信 华莱士独立构想同一原理，1858年来稿几乎与达尔文一致，令达尔文担心自己的首创地位。
- 折中 莱尔、胡克安排两人成果于1858年7月同场宣读；两位作者都未出席，华莱士也未参与商定办法。
- 反响 论文发表初期反响冷淡；1859年《物种起源》引起轰动，首印1250册已被书商全部订购。
- 认可 华莱士1858年10月事后致信认可安排；1889年又以《达尔文主义》阐述自然选择，未用自己名字命名理论。

参考答案　1 amassing 2 priority 3 muted 4 cordial

入侵澳洲的毒蟾

The Toad That Invaded Australia



引进的灭虫帮手，为什么成了生态难题？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Greyback cane beetles, whose larvae devour roots, were plaguing lucrative sugar plantations in Queensland, north-eastern Australia. In June 1935 Queensland imported 102 cane toads, bulky amphibians native to the Americas, from Hawaii. That August about 2,400 of their young were released. Biological control, setting one species against a pest, had an ambitious new recruit.

The hasty plan met opposition. In November 1935 an insect expert, Walter Froggatt, persuaded officials to halt releases; after Queensland's lobbying, they resumed in September 1936. The scheme flopped: the toads barely dented beetle numbers, perhaps because the insects stayed high up on cane stalks.

The toads thrived instead. With few foes and fertile females spawning thousands of eggs, they multiplied unchecked. Their invasion frontier, creeping about ten kilometres a year at first, accelerated to about 50 a year in the tropical north. In February 2009 the first known toad breached Western Australia's border.

Toxin glands in their skin make them a deadly meal. Numbers of goannas (large lizards) and northern quolls (small marsupial hunters) have slumped in places; some local quoll populations were wiped out. The toads have wreaked havoc, but the damage is uneven; some species have prospered.

Biological control can triumph. From 1926 the cactus moth, Cactoblastis, cleared most prickly pear, an invasive cactus, from Queensland and New South Wales within about a decade. Tested first, its larvae starved rather than eat fifteen other plants; the toads faced no such test. Under an October 1984 law, recruits now face stringent scrutiny. The blunder was born of haste and sealed by lobbying that overrode a warning: an untested ally, once loose, may never be recalled.

中文译文

在澳大利亚东北部昆士兰州，灰背蔗甲虫的幼虫啃食根系，困扰着利润丰厚的甘蔗种植业。1935年6月，当地从夏威夷引进102只甘蔗蟾蜍。这些体型笨重的两栖动物原产美洲。同年8月，约2400只繁育出的幼蟾被放到野外。用一种生物控制另一种害虫的生物防治，就这样迎来了一位雄心勃勃的“新兵”。

这项草率的计划遭到反对。1935年11月，昆虫专家沃尔特·弗罗格特说服官员暂停释放，但昆士兰州游说后，释放又在1936年9月恢复。计划最终失败：蟾蜍几乎没有减少甲虫数量，可能因为这些昆虫待在高高的甘蔗茎秆上。

蟾蜍自身反而繁盛起来。它们天敌很少，雌蟾能产下数千枚卵，于是数量不受控制地增加。入侵前沿最初每年推进约10公里，在热带北部加速到每年约50公里。2009年2月，已知的第一只蟾蜍越过了西澳大利亚州边界。

蟾蜍皮肤里有毒腺，捕食者吃下它们可能送命。一些地区的巨蜥和北袋鼬数量大幅下降，某些地方的袋鼬种群甚至消失了。巨蜥是大型蜥蜴，北袋鼬则是小型有袋类捕食者。蟾蜍造成了严重生态破坏，但影响并不均匀，一些物种反而兴盛起来。

生物防治也有成功的例子。自1926年起，仙人掌螟在约十年间清除了昆士兰州和新南威尔士州大部分入侵的仙人掌属植物。它的幼虫事先经过检验，宁可饿死也不吃另外15种植物；蟾蜍却没有接受过这样的测试。根据1984年10月通过的法律，如今引入生物防治物种需要严格审查。这次失误始于仓促，又因游说压过警告而成为定局：一个未经检验的“盟友”，一旦放出，就可能再也召不回来。

原文只说甲虫在高处“可能”解释防治失败；生态损害也并非对所有物种相同。

核心词汇

先记这22项。连着下方的原文短语一起回想。

larva n.
幼虫
whose larvae devour roots

devour v.
大量啃食
larvae devour roots

lucrative adj.
利润丰厚的
lucrative sugar plantations

amphibian n.
两栖动物
bulky amphibians

native to 形容词短语
原产于
native to the Americas

species n.
物种
setting one species against a pest

pest n.
害虫；有害生物
one species against a pest

hasty adj.
草率仓促的
The hasty plan met opposition

opposition n.
反对
The hasty plan met opposition

persuade v.
说服
persuaded officials to halt releases

lobbying n.
游说
after Queensland's lobbying

resume v.
恢复；重新开始
they resumed in September 1936

thrive v.
繁盛
The toads thrived instead

unchecked adj.
未受遏制的
they multiplied unchecked

invasion n.
入侵
Their invasion frontier

accelerate v.
加速
accelerated to about 50 a year

toxin n.
毒素
Toxin glands in their skin

wipe out 动词短语
彻底消灭
quoll populations were wiped out

invasive adj.
入侵性的
an invasive cactus

stringent adj.
严格的
stringent scrutiny

scrutiny n.
仔细审查
stringent scrutiny

override v.
压过；推翻
lobbying that overrode a warning

文中词汇与搭配

补充词汇共40项。保留每个表达在本文中的意思。

beetle n.
甲虫

plague v.
困扰

plantation n.
种植园

import v.
引进

bulky adj.
体积庞大笨重的

release v.
释放

biological adj.
生物的

ambitious adj.
雄心勃勃的

recruit n.
新成员；新兵

halt v.
停止

scheme n.
计划

flop v.
彻底失败

dent v.
削减；产生影响

stalk n.
茎秆

foe n.
敌手；此处指天敌

fertile adj.
能繁殖的

spawn v.
产卵

multiply v.
大量繁殖

frontier n.
前沿

tropical adj.
热带的

文中词汇与搭配 续

补充词汇共40项。保留每个表达在本文中的意思。

breach v.
突破；越过

gland n.
腺体

lizard n.
蜥蜴

marsupial adj.
有袋类的

slump v.
锐减

wreak v.
造成（破坏）

havoc n.
严重破坏

uneven adj.
不均等的

prosper v.
兴盛

triumph v.
大获成功

moth n.
蛾

prickly adj.
多刺的

cactus n.
仙人掌

starve v.
饿死

blunder n.
严重错误

haste n.
仓促

untested adj.
未经检验的

ally n.
盟友

loose adj.
被放出的；不受约束的

recall v.
召回

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Greyback cane beetles, whose larvae devour roots, were plaguing _____ sugar plantations in Queensland, north-eastern Australia.

02 The hasty plan met _____.

03 With few foes and fertile females spawning thousands of eggs, they multiplied _____.

04 Under an October 1984 law, recruits now face stringent _____.

文章主线

- 引进 1935年，澳大利亚昆士兰引进102只甘蔗蟾蜍，放出约2,400只后代，想控制啃食甘蔗根系的甲虫。
- 警告 昆虫专家曾促成暂停释放，地方游说却使1936年恢复；蟾蜍释放前缺少食性检验，警告最终被压过。
- 失控 甲虫几乎没减少，或因它们待在高处；蟾蜍却天敌少、产卵多，在热带北部每年扩散约50公里。
- 伤害 蟾蜍的皮肤毒素使巨蜥、北袋鼬在部分地区锐减，某些袋鼬种群消失；也有物种反而兴盛。
- 对照 仙人掌螟经检验不吃另15种植物，后来成功控制入侵仙人掌；1984年立法要求严格审查生物防治引种。

参考答案　1 lucrative 2 opposition 3 unchecked 4 scrutiny

从一道划痕到天花绝迹

From a Scratch to a World Without Smallpox



詹纳不是第一人，贡献究竟在哪里?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Smallpox, a contagious affliction, killed about three in ten of its sufferers and an estimated 300 million in the 20th century. An early safeguard, deliberately implanting matter from its sores in the healthy, was practised in China by the 16th century and reached England in 1721. The procedure itself could cause grave sickness or death, if less frequently than natural smallpox.

Rustic lore held that milkmaids with cowpox, a mild ailment of livestock, were immune to smallpox. In 1774 Benjamin Jesty, a Dorset farmer, employed cowpox to protect his wife and two sons. On 14 May 1796 Edward Jenner, a provincial practitioner, inserted fluid from a milkmaid's cowpox sore into a scratch on eight-year-old James Phipps.

On 1 July 1796 Jenner exposed the boy to smallpox; no symptoms ensued. In 1797 the Royal Society declined to publish his paper, judging his evidence unconvincing. In June 1798 he privately funded his book, the Inquiry, with supplementary cases. The word vaccine derives from the Latin vacca, cow.

Jenner was not the idea's originator. His contribution was a test, with results accessible in print, so other doctors could replicate and disseminate the technique. By today's standards, experimentation on a child without informed consent would be unethical. Jenner died on 26 January 1823, with smallpox far from eliminated.

In 1967 the WHO launched an intensified, concerted campaign to eradicate smallpox. The last naturally occurring case was diagnosed in Somalia on 26 October 1977. On 8 May 1980 the World Health Assembly formally declared the world free of smallpox, the only human disease ever wiped out. That pronouncement came almost 184 years after Phipps was vaccinated.

中文译文

天花是一种传染病，约有三成患者因此死亡；仅在20世纪，估计就夺走了3亿人的生命。一种早期预防办法，是把取自天花疮疹的物质接种给健康人。这种做法到16世纪已在中国使用，1721年传到英国。接种本身也可能造成重病或死亡，只是比自然感染天花时少见。

乡间相传，挤奶女工得过牛痘后，就会对天花免疫；牛痘是牲畜的一种轻微疾病。1774年，多塞特郡农夫本杰明·杰斯蒂就曾用牛痘来保护妻子和两个儿子。1796年5月14日，乡村医生爱德华·詹纳把取自一名挤奶女工牛痘疮的液体，接种到8岁男孩詹姆斯·菲普斯身上的一道划痕中。

1796年7月1日，詹纳让男孩接触天花，男孩没有出现症状。1797年，皇家学会认为证据缺乏说服力，拒绝发表他的论文。1798年6月，他自费出版《探究》，补充了更多病例。英语中表示疫苗的vaccine一词，来自拉丁语vacca，意思是牛。

詹纳不是这个想法的首创者。他的贡献在于开展试验，并把结果公开出版，让其他医生能够重复验证、传播这项技术。按今天的标准，在没有知情同意的情况下拿孩子做实验，是不合伦理的。詹纳于1823年1月26日去世，那时天花还远未被消除。

1967年，世界卫生组织发起更强有力的协同行动，决心根除天花。最后一例自然发生的天花病例，于1977年10月26日在索马里确诊。1980年5月8日，世界卫生大会正式宣布全球摆脱天花；它也是迄今唯一被彻底根除的人类疾病。此时，距菲普斯接种已过去近184年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

contagious adj.
有传染性的
Smallpox, a contagious affliction,

affliction n.
疾病；苦难
Smallpox, a contagious affliction,

safeguard n.
防护措施
safeguard, deliberately implanting

implant v.
植入；接种
safeguard, deliberately implanting

procedure n.
做法；操作程序
The procedure itself could cause grave

grave adj.
严重的
itself could cause grave sickness or

immune to adj.phr.
对……免疫的
of livestock, were immune to smallpox.

practitioner n.
执业医师
Jenner, a provincial practitioner,

ensue v.
随后发生
no symptoms ensued.

supplementary adj.
补充的
with supplementary cases.

vaccine n.
疫苗
The word vaccine derives from the

originator n.
首创者
Jenner was not the idea's originator.

accessible adj.
可以获得或查阅的
was a test, with results accessible in

replicate v.
重复验证；复制
so other doctors could replicate and

disseminate v.
传播；散布
could replicate and disseminate the

informed adj.
知情的
on a child without informed consent

consent n.
同意
on a child without informed consent

unethical adj.
不合伦理的
would be unethical.

concerted adj.
协同一致的
an intensified, concerted campaign to

eradicate v.
根除
campaign to eradicate smallpox.

diagnose v.
确诊
was diagnosed in Somalia on 26 October

pronouncement n.
正式宣告
That pronouncement came almost 184

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

smallpox n.
天花

sufferer n.
患者

deliberately adv.
有意地

matter n.
物质；此处指痘疮取材

sore n.
疮；疮疹

practise v.
施行；采用

sickness n.
疾病

frequently adv.
经常；频繁地

rustic adj.
乡间的

lore n.
民间传说；传统知识

mild adj.
轻微的

ailment n.
病症；小病

livestock n.
牲畜

employ v.
使用；采用

provincial adj.
地方的；外省的

insert v.
放入；植入

fluid n.
液体

scratch n.
划痕

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

expose to phr.v.
与to连用：使接触

to prep.
标示所接触的对象

symptom n.
症状

publish v.
发表；出版

unconvincing adj.
缺乏说服力的

privately adv.
由私人出资地；自费地

fund v.
出资；资助

derive from phr.v.
源于

Latin n.
拉丁语

technique n.
技术；方法

experimentation n.
试验；实验

eliminate v.
消除

intensified adj.
加强了的

naturally adv.
自然地

occur v.
发生

formally adv.
正式地

wipe out phr.v.
彻底消灭

vaccinate v.
接种疫苗

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Smallpox, a _____ affliction,

02 no symptoms _____.

03 so other doctors could _____ and disseminate the technique.

04 an intensified, concerted campaign to _____ smallpox.

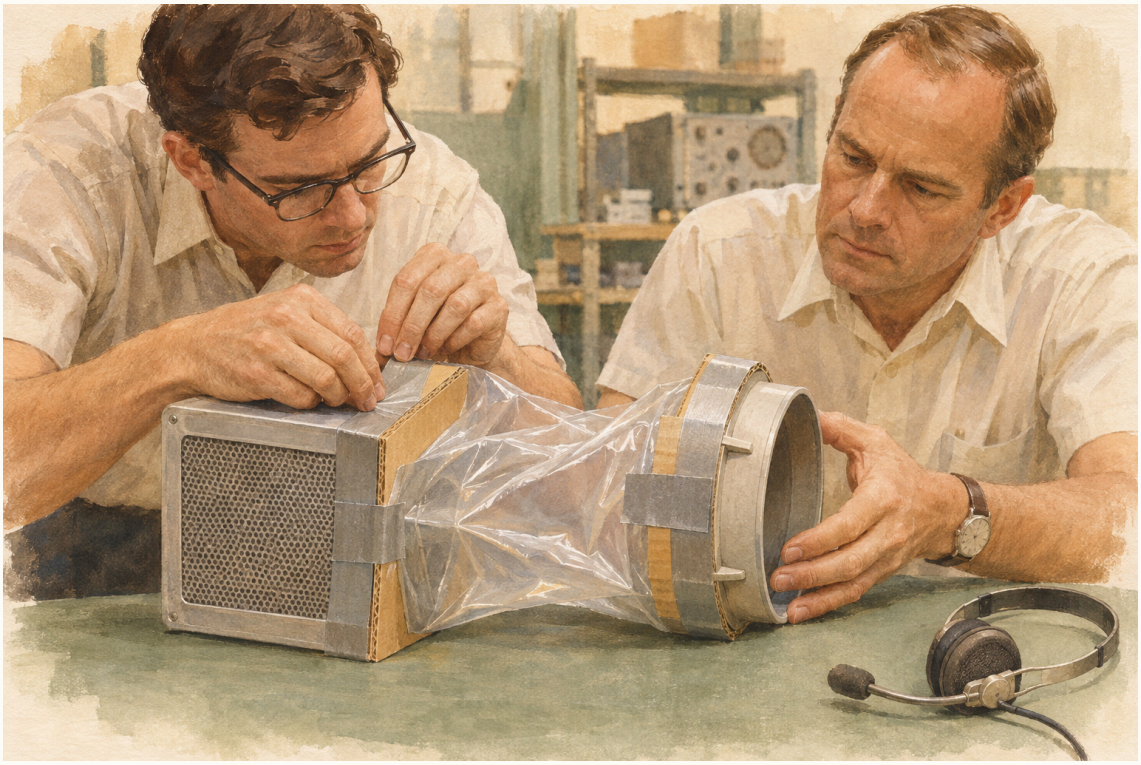
文章主线

- 背景 天花约致三成患者死亡；早期人痘术把天花疮中物质接种给健康人作预防，风险虽低于自然染病，仍可能致重病或死亡。
- 先例 乡间传牛痘可防天花；农夫杰斯蒂1774年已用牛痘保护家人，詹纳并非首创。
- 试验 詹纳1796年给男孩种牛痘，再让他接触天花而未发病；无知情同意的儿童试验按今天标准不合伦理。
- 推广 皇家学会嫌证据不足拒稿后，詹纳自费公布试验和补充病例，使其他医生能够重复验证、传播技术。
- 根除 世卫1967年强化行动，1977年出现最后自然病例；1980年世界卫生大会宣布根除天花，迄今唯一被根除的人类疾病。

参考答案　1 contagious 2 ensued 3 replicate 4 eradicate

阿波罗13号：成功的失败

Apollo 13: The Successful Failure



方形滤芯怎样接进圆形插槽，救回三名宇航员？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On April 11th 1970 NASA launched Apollo 13, meant for the third crewed lunar landing. On April 13th, some 200,000 miles out, an oxygen tank exploded in the service module, home to most of the ship's oxygen and power. The blast crippled the spacecraft; the landing was aborted.

The astronauts took refuge in the lunar module, Aquarius, built to support two men for about two days. Using its engine, the crew steered the craft onto a trajectory around the Moon, whose gravity swung it home. They rationed water, conserved power, and shivered in a bitterly cold cabin.

A quieter hazard emerged: with a third man aboard, carbon dioxide was accumulating. The command module, the crew's capsule, used square filter cartridges; Aquarius's sockets were round. From plastic bags, cardboard, and tape on board, ground engineers devised an adapter, then talked the crew through assembling it. On April 17th the capsule splashed down in the Pacific.

Yet the fault had been built in on Earth. A NASA review board traced the explosion to the tank's thermostat switches, designed for the ship's 28-volt supply. In 1965 the spacecraft's builder rewrote the specification to let the heaters run on 65-volt launch-site power; the switches were never upgraded, a mismatch neither that contractor nor NASA detected. Weeks before launch, to drain the stubborn tank, a heater ran about eight hours. The switches probably malfunctioned; unchecked heat scorched the insulation on the tank's wiring; in flight, that insulation ignited.

NASA called Apollo 13 a "successful failure". The paradox holds: a flight undone by a five-year-old engineering oversight was salvaged within four days by ingenuity on the ground.

中文译文

1970年4月11日，NASA发射阿波罗13号，原计划完成第三次载人登月。4月13日，飞船已飞出约20万英里，服务舱内一个氧气罐爆炸。飞船大部分氧气和电力都由服务舱提供；爆炸使航天器严重受损，登月任务被取消。

宇航员转入“水瓶座”号登月舱避难，而它原本只能维持两个人生存约两天。机组用登月舱发动机调整航向，使飞船沿绕月轨迹飞行，再借月球引力转向地球。他们限量用水、节约电力，在严寒的舱内瑟瑟发抖。

另一种不易察觉的危险出现了：舱内多了第三个人，二氧化碳不断积聚。机组乘坐的指令舱使用方形滤芯，水瓶座号的插槽却是圆的。地面工程师利用飞船上现有的塑料袋、硬纸板和胶带设计转接装置，再通过通话一步步指导机组组装。4月17日，指令舱在太平洋溅落。

然而，隐患早在地球上就已埋下。NASA审查委员会将爆炸原因追溯到罐内的温控开关：它们按飞船28伏电源设计。1965年，航天器承建商修改技术规格，允许加热器使用发射场65伏电源，开关却始终没有升级。承包商和NASA都未发现这种不匹配。发射前几周，为排空那个难以排空的储罐，一台加热器连续运行约八小时。温控开关很可能失灵，失去控制的高温烧焦罐内电线的绝缘层；飞行途中，绝缘材料起火。

NASA把阿波罗13号称为“成功的失败”。这看似矛盾的评价确实成立：一次飞行毁于五年前的工程疏忽，却在四天内靠地面人员的巧思得以挽救。

核心词汇

先记这22项。连着下方的原文短语一起回想。

cripple v.
使严重受损
The blast crippled the spacecraft

abort v.
中止；取消
the landing was aborted

refuge n.
避难；庇护
took refuge in the lunar module

trajectory n.
轨迹；飞行路线
a trajectory around the Moon

ration v.
限量供应；限量使用
They rationed water

hazard n.
危险；隐患
A quieter hazard emerged

emerge v.
出现
A quieter hazard emerged

accumulate v.
积聚
carbon dioxide was accumulating

devise v.
设计出；想出
ground engineers devised an adapter

adapter n.
转接装置
devised an adapter

assemble v.
组装
through assembling it

fault n.
缺陷；故障
the fault had been built in on Earth

thermostat n.
温控器
the tank's thermostat switches

specification n.
技术规格
rewrote the specification

mismatch n.
不匹配
a mismatch

drain v.
排空
to drain the stubborn tank

malfunction v.
失灵；发生故障
The switches probably malfunctioned

insulation n.
绝缘材料
the insulation on the tank's wiring

paradox n.
看似矛盾却成立的事
The paradox holds

oversight n.
疏忽
a five-year-old engineering oversight

salvage v.
挽救
was salvaged within four days

ingenuity n.
巧思；足智多谋
by ingenuity on the ground

文中词汇与搭配

补充词汇共52项。保留每个表达在本文中的意思。

launch v.
发射

crewed adj.
载人的

lunar landing 名词短语
登月

oxygen tank 名词短语
氧气罐

explode v.
爆炸

service module 名词短语
服务舱

blast n.
爆炸；爆炸冲击

spacecraft n.
航天器

astronaut n.
宇航员

take v.
采取；与refuge连用指避难

in prep.
在……中；在此引出避难处

lunar module 名词短语
登月舱

engine n.
发动机

steer v.
操纵；引导航向

craft n.
航天器

onto prep.
进入；在此指进入某条轨迹

swing v.
使转向；甩回

conserve power 动词短语
节约电力

shiver v.
因寒冷而发抖

bitterly adv.
极其；非常

cabin n.
舱室

carbon dioxide 名词短语
二氧化碳

capsule n.
太空舱；返回舱

filter n.
过滤器

cartridge n.
可替换的滤芯

socket n.
插槽；接口

文中词汇与搭配 续

补充词汇共52项。保留每个表达在本文中的意思。

cardboard n.
硬纸板

tape n.
胶带

talk v.
口头指导；与through连用

through prep.
贯穿整个过程；与talk连用

splash down 动词短语
在水上溅落

build in 动词短语
预先嵌入；埋下

review board 名词短语
审查委员会

trace v.
追溯；查明来源

explosion n.
爆炸

to prep.
用于引出追溯到的原因

switch n.
开关

volt n.
伏特；电压单位

builder n.
承建商

rewrite v.
改写；修改

heater n.
加热器

run on 动词短语
靠某种能源运行

upgrade v.
升级

contractor n.
承包商

detect v.
发现；查出

stubborn adj.
难处理的；此处指难排空的

heater n.
加热器

unchecked adj.
未受控制的

scorch v.
烧焦

wiring n.
电线；线路

ignite v.
起火；燃烧

be undone by 动词短语
被……毁掉；因……而失败

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The blast crippled the spacecraft; the landing was _____.

02 They _____ water, conserved power, and shivered in a bitterly cold cabin.

03 A quieter _____ emerged: with a third man aboard, carbon dioxide was accumulating.

04 a flight undone by a five-year-old engineering _____

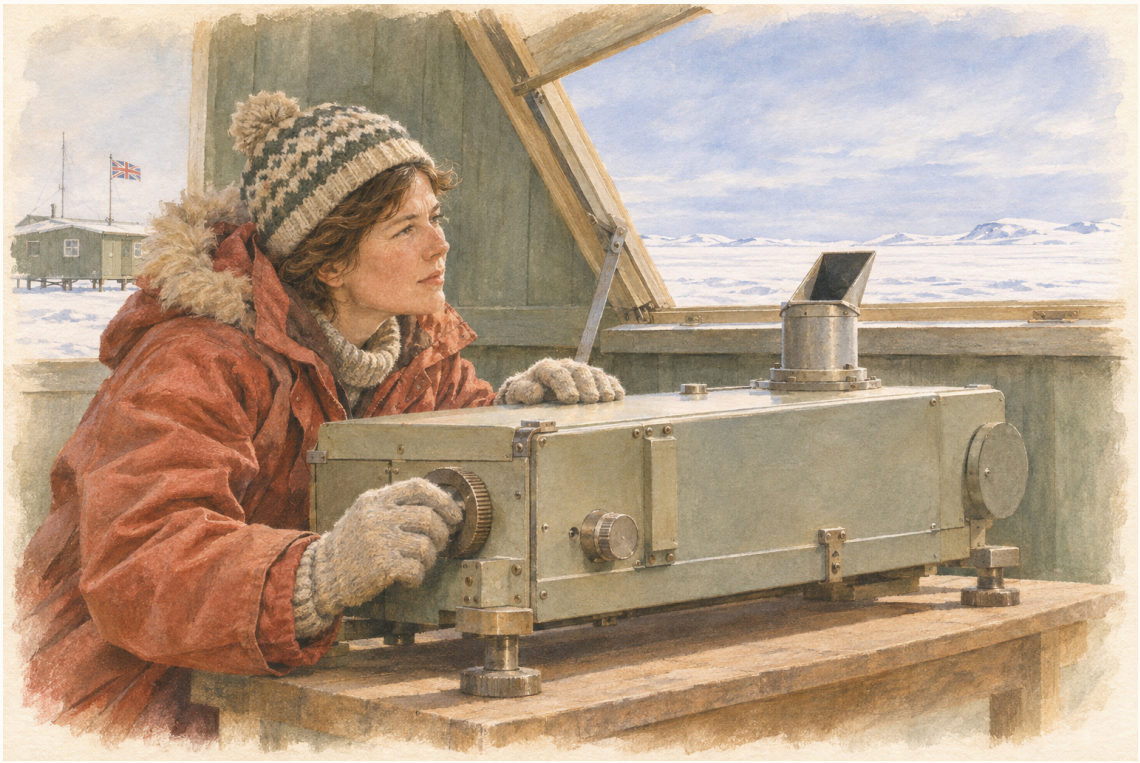
文章主线

- 事故 1970年，阿波罗13号氧气罐爆炸，登月取消，三名宇航员转入原本只供两人约两天的登月舱。
- 险情 机组缺水缺电、忍受严寒，二氧化碳不断积聚，清除它的方形滤芯却装不进圆形插槽。
- 解围 地面工程师用船上的塑料袋、纸板和胶带设计转接器，指导机组组装，让滤芯接上圆形插槽。
- 祸根 发射前，加热器用65伏地面电源，温控开关仍按28伏设计；开关疑失灵，约八小时加热损伤线路绝缘层，飞行中起火。
- 返航 机组用登月舱发动机调整绕月路线，借月球引力返航；爆炸四天后，三人安全溅落太平洋。

参考答案　1 aborted 2 rationed 3 hazard 4 oversight

拯救天空

Saving the Sky



臭氧层的低值，为什么被软件怀疑?

READ
纯净阅读

CHECK
中文核对

LEARN
词汇记忆

RECALL
回想巩固

英文原文

A layer of ozone shields Earth from much of the sun's ultraviolet radiation. In June 1974 Mario Molina and Sherwood Rowland, chemists in California, warned in Nature of a menace in fridges and spray cans. Chlorofluorocarbons (CFCs) are so inert that they drift intact into the stratosphere, high in the atmosphere, where they can liberate chlorine, a catalyst that destroys ozone molecule after molecule. The pair foresaw gradual erosion.

The alarm, when it sounded, was abrupt. In May 1985 Joseph Farman and colleagues at the British Antarctic Survey reported that ozone over Halley Bay, their Antarctic station, had plummeted each southern spring since the mid-1970s. Their humble Dobson spectrophotometer gauged ozone from the ground. The staff were persistent, reading it in a painstaking routine of manual observation.

Yet NASA's Nimbus-7 satellite, a sophisticated eye in orbit, had recorded the same lows. Its software flagged them as dubious, since ozone that low had never been reliably measured. NASA says it was still verifying them when Farman published. The sensor had worked; expectation had not.

Remarkably, the response was swift. In September 1987 diplomats signed the Montreal Protocol, a binding accord to curb CFCs. Amendments tightened it: in June 1990 affluent countries pledged to phase out CFCs by 2000. In September 2009 the Protocol and its parent Vienna Convention became the UN's first treaties to win universal ratification.

In January 2023 a UN-backed assessment forecast recovery to 1980 levels around 2066 over Antarctica, 2045 over the Arctic, and 2040 elsewhere. Healing will take decades. The sky's rescue began with diligent observers who trusted that a baffling number was no glitch, and published it.

中文译文

臭氧层为地球挡住了太阳的大量紫外线。1974年6月，在加州工作的化学家马里奥·莫利纳和舍伍德·罗兰在《自然》发出警告：冰箱和喷雾罐里藏着一种威胁。氯氟烃，也就是CFCs，性质很不活泼，能完整地飘入高空的平流层，在那里释放氯。氯是一种催化剂，会接连破坏臭氧分子。两人预料到的是逐步发生的损耗。

真正的警报却来得很突然。1985年5月，约瑟夫·法曼和英国南极调查局的同事报告：自20世纪70年代中期起，他们南极哈雷湾考察站上空的臭氧，每逢南半球春季都会骤降。他们使用一台不起眼的多布森分光光度计，从地面测量臭氧。工作人员持之以恒，遵循细致的固定流程，靠人工观测读数。

然而，美国国家航空航天局的雨云7号卫星，这只精密的太空眼睛，也已测到同样的低值。由于此前从未可靠地测到这么低的臭氧含量，软件把这些读数标成可疑。NASA表示，法曼发表论文时，他们仍在核实数据。传感器没有出错，出错的是人的预期。

各国的反应迅速得令人瞩目。1987年9月，外交官签署了具有约束力的《蒙特利尔议定书》，限制氯氟烃的使用。之后的修正案进一步收紧管控：1990年6月，富裕国家承诺在2000年前逐步淘汰氯氟烃。2009年9月，议定书及其所依据的《维也纳公约》，成为联合国首批获得所有国家批准的条约。

2023年1月，一份由联合国支持的评估报告预测，臭氧层恢复到1980年水平的时间，南极上空约为2066年，北极约为2045年，其他地区约为2040年。愈合仍需数十年。拯救天空的起点，是一群勤勉的观测者：他们相信，那组令人费解的数字不是故障，于是将其发表。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ozone n.
臭氧
layer of ozone shields Earth from much

menace n.
威胁
of a menace in fridges and spray cans.

inert adj.
化学性质不活泼的
inert that they drift intact into the

intact adj.
完好无损的
inert that they drift intact into the

liberate v.
释放
where they can liberate chlorine,

catalyst n.
催化剂
catalyst that destroys ozone molecule

erosion n.
侵蚀；逐步损耗
The pair foresaw gradual erosion.

abrupt adj.
突然的
alarm, when it sounded, was abrupt.

plummet v.
骤降
plummeted each southern spring since

persistent adj.
坚持不懈的
The staff were persistent, reading it

painstaking adj.
细致认真的
it in a painstaking routine of manual

sophisticated adj.
精密复杂的
a sophisticated eye in orbit,

dubious adj.
可疑的
Its software flagged them as dubious,

verify v.
核实
was still verifying them when Farman

binding adj.
有约束力的
a binding accord to curb CFCs.

accord n.
协定
a binding accord to curb CFCs.

curb v.
限制；遏制
a binding accord to curb CFCs.

pledge v.
承诺
June 1990 affluent countries pledged

phase out phr.v.
逐步淘汰
countries pledged to phase out CFCs by

ratification n.
正式批准
to win universal ratification.

assessment n.
评估
2023 a UN-backed assessment forecast

baffling adj.
令人费解的
a baffling number was no glitch, and

文中词汇与搭配

补充词汇共40项。保留每个表达在本文中的意思。

layer n.
层

shield from phr.v.
与from连用：保护；遮挡

from prep.
标示所防护的危害

ultraviolet adj.
紫外线的

radiation n.
辐射

spray n.
喷雾

atmosphere n.
大气层

chlorine n.
氯

molecule n.
分子

gradual adj.
逐渐的

alarm n.
警报

Antarctic adj.
南极的

humble adj.
不起眼的；简朴的

gauge v.
测量

routine n.
固定流程；例行程序

manual adj.
人工的；手工的

observation n.
观测

satellite n.
卫星

in orbit prep.phr.
在轨道上运行

flag as phr.v.
与as连用：标记为

文中词汇与搭配 续

补充词汇共40项。保留每个表达在本文中的意思。

as prep.
标示被认定的类别

sensor n.
传感器

expectation n.
预期

remarkably adv.
引人注目地

swift adj.
迅速的

diplomat n.
外交官

protocol n.
议定书

amendment n.
修正案

tighten v.
收紧；加强

affluent adj.
富裕的

treaty n.
条约

universal adj.
全体的；覆盖所有国家的

forecast v.
预测

recovery n.
恢复

Antarctica proper n.
南极洲

Arctic proper n.
北极地区

healing n.
愈合；修复

rescue n.
拯救

diligent adj.
勤勉的

glitch n.
小故障

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 a _____ that destroys ozone molecule after molecule.

02 Its software flagged them as _____,

03 a _____ accord to curb CFCs.

04 affluent countries pledged to _____ CFCs by 2000.

文章主线

- 预警 臭氧挡紫外线。1974年，化学家警告：冰箱和喷罐的氯氟烃可进入平流层，释放氯持续破坏臭氧。
- 发现 1985年南极地面观测报告：哈雷湾臭氧自70年代中期起，每逢南半球春季便骤降。
- 误判 卫星同样测到低值，软件因罕见而标为可疑；NASA称当时尚在核实，是预期出了错，并非仪器失灵。
- 合作 各国1987年签署《蒙特利尔议定书》，1990年富国承诺2000年前淘汰氯氟烃，议定书2009年获所有国家批准。
- 恢复 2023年评估预测恢复至1980年水平：其他地区约2040年、北极约2045年、南极约2066年。

参考答案　1 catalyst 2 dubious 3 binding 4 phase out

写给外星人的一张唱片

The Record Humanity Threw into the Stars



如果有人找到这张唱片，会认识地球吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In late summer 1977, NASA fastened a disc wrought from gilded copper, 30 centimetres wide, to each Voyager spaceship before takeoff. Nobody could foresee who, if anyone, would play one.

Advisers under Carl Sagan of Cornell distilled our earthly abode into this compendium. Voices greet the listener in 55 languages, followed by the rumble of thunder, the whistle of wind, whales, footsteps, laughter, and a rhythmic heartbeat. About 90 minutes of eclectic music cover a spectrum of cultures and eras, notably Bach and Chuck Berry's Johnny B. Goode.

An array of 115 images, encoded in analogue form, sits beside printed words from two statesmen, Jimmy Carter and Kurt Waldheim. The sleeve gives guidance on playing the disc, plus a diagram fixing the Sun with pulsars as beacons.

On 14 February 1990, at Sagan's request, Voyager 1 pivoted its camera for a glance backwards across about six billion kilometres. Sixty frames formed a panorama of the solar system, a mosaic in which Earth was a glimmer about an eighth of a pixel across. They reached Earth's receivers between March and 1 May, each signal taking five and a half hours.

In a 1994 book, Sagan offered that pale dot as a perspective on our small and ephemeral place amid cosmic vastness.

Voyager 1 crossed into interstellar space in August 2012, the first human-made object there, and Voyager 2 followed in 2018. Both work on with a longevity unmatched at NASA, but depletion of their power should silence them within a few years.

The Voyagers' discs will glide on outwards through the void as messengers with a map leading back to the Sun.

中文译文

1977年夏末，美国航天局在两艘旅行者号探测器起飞前，各固定上一张直径30厘米的镀金铜唱片。谁也不知道，究竟会不会有谁、又会是谁来播放它们。

康奈尔大学卡尔·萨根带领的顾问团队，把地球家园浓缩成这份汇编。人声用55种语言向听者问候，接着是雷声、风声、鲸声、脚步声、笑声和有节奏的心跳。约90分钟的音乐包容了不同文化和时代，其中包括巴赫的作品，以及查克·贝里的《Johnny B. Goode》。

唱片中还有以模拟方式编码的115幅图像，并附有两位政治家吉米·卡特和库尔特·瓦尔德海姆的书面致辞。封套上给出了播放方法，以及一幅用脉冲星作信标、标定太阳位置的示意图。

1990年2月14日，应萨根的请求，旅行者1号转动相机，从约60亿公里外回望。60帧画面组成太阳系全景；地球在其中只是一点微光，宽度约为八分之一个像素。这些画面在3月至5月1日之间传回地球，信号单程需要五个半小时。

在1994年的一本书中，萨根借那颗暗淡的小点，帮助人们重新审视自己在浩瀚宇宙中渺小、短暂的位置。

旅行者1号于2012年8月进入星际空间，成为首个到达那里的人工物体；旅行者2号于2018年跟进。两者仍在工作，寿命在美国航天局的任务中无可匹敌；但随着电力耗尽，预计几年内就会沉寂。

旅行者号携带的唱片仍将继续向外穿越虚空，作为来自人类的信使，带着一幅指回太阳的地图。

文中的信号需五个半小时，是从当时旅行者1号的位置到地球的单程时间。

核心词汇

先记这22项。连着下方的原文短语一起回想。

fasten v.
固定；系牢
NASA fastened a disc

wrought adj.
制成的；加工成形的
a disc wrought from gilded copper

gilded adj.
镀金的
gilded copper

foresee v.
预见
Nobody could foresee

distil v.
提炼；浓缩
distilled our earthly abode

abode n.
住所；家园
our earthly abode

compendium n.
汇编
into this compendium

eclectic adj.
兼收并蓄的
About 90 minutes of eclectic music

encode v.
编码
encoded in analogue form

guidance n.
指引
guidance on playing the disc

beacon n.
信标
with pulsars as beacons

pivot v.
转动
Voyager 1 pivoted its camera

panorama n.
全景
a panorama of the solar system

mosaic n.
拼接图
a mosaic in which Earth was a glimmer

pixel n.
像素
an eighth of a pixel across

signal n.
信号
signal taking five and a half hours

perspective n.
看待问题的视角
as a perspective

ephemeral adj.
短暂的
our small and ephemeral place

interstellar adj.
恒星之间的
crossed into interstellar space

longevity n.
长寿；长久持续
a longevity unmatched at NASA

unmatched adj.
无可匹敌的
a longevity unmatched at NASA

depletion n.
耗竭
depletion of their power

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

disc n.
唱片；圆盘

copper n.
铜

centimetre n.
厘米

spaceship n.
宇宙飞船；航天器

takeoff n.
起飞

adviser n.
顾问

earthly adj.
地球上的

greet v.
向……问候

listener n.
听者

rumble n.
隆隆声

thunder n.
雷声

whistle n.
呼啸声

laughter n.
笑声

rhythmic adj.
有节奏的

heartbeat n.
心跳

a spectrum of 名词短语
一系列；多种多样的

era n.
时代

notably adv.
尤其

an array of 名词短语
一系列

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

statesman n.
政治家

sleeve n.
唱片封套

diagram n.
示意图

at prep.
应（某人的请求）

request n.
请求

glance n.
一瞥

backwards adv.
向后

frame n.
画面；帧

glimmer n.
微光

across adv.
从一侧到另一侧；宽

receiver n.
接收设备

pale adj.
暗淡的

cosmic adj.
宇宙的

vastness n.
浩瀚；辽阔

human-made adj.
人造的

glide on 动词短语
继续滑行

outwards adv.
向外

void n.
虚空

messenger n.
信使

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Nobody could _____ who, if anyone, would play one.

02 Advisers under Carl Sagan of Cornell distilled our earthly abode into this _____.

03 An array of 115 images, _____ in analogue form, sits beside printed words from two statesmen, Jimmy Carter and Kurt Waldheim.

04 In a 1994 book, Sagan offered that pale dot as a perspective on our small and _____ place amid cosmic vastness.

文章主线

- 出发 1977年，两艘旅行者号探测器各携一张镀金铜唱片，萨根团队以此介绍地球；会不会有人播放，仍是未知。
- 内容 唱片有55种语言的问候、自然和生活声响、约90分钟音乐与115幅图像，还附时任美国总统与联合国秘书长的文字致辞。
- 指引 封套附播放指引，并把脉冲星当作定位信标，标出太阳的位置，告诉潜在听者唱片来自哪里。
- 回望 1990年，旅行者1号在约60亿公里外回拍，地球仅约八分之一像素；萨根借此审视人类在宇宙中的渺小短暂。
- 远行 旅行者1号2012年成为首个进入星际空间的人造物，2号2018年跟进；电力终会耗尽，唱片仍将远行。

参考答案　1 foresee 2 compendium 3 encoded 4 ephemeral